

The Constants of Nature

$\varepsilon_{\text{h}} \colon \text{K} = \frac{\text{kelvin}}{\text{K}_1} \left(\frac{T_{\text{p}}}{T_{\text{p}} m_{\text{p}}} \right) \left(1 - 14 C_{\text{U}}^{-4} \mathscr{K}_{\text{p}} \right) \mathfrak{E}$	$m_{\text{de}} = \frac{3\text{s}}{\mathscr{K}_{\text{3}} \mathscr{K}_{\text{4}}} \left(\frac{16}{4\pi} \left(m_{\text{n}} + m_{\text{e}} \right) \right) \left(1 + \frac{Im(\omega_{\text{1}})}{1 + \mathscr{K}_2} \left(\frac{L_2}{L_1} \mathscr{K}_{\text{e}}^{-3} \right) \right) \mathfrak{E}$	$\varepsilon_{\text{p}} = \frac{\text{joule}}{\text{GeV}} \left(\frac{t_{\text{p}}^2 m_{\text{p}}}{t_{\text{p}}^2} \right) \left(1 - \frac{1}{2} Re \left(i^{i'''} \right) \mathscr{K}_{\text{3}} \mathscr{K}_{\text{4}} \mathfrak{E} \right)$	$\frac{\mu_{\text{h}}}{hc} = \frac{\mathscr{K}_{\text{1}}}{4\pi} \left(\frac{t_{\text{p}} q_{\text{p}}}{t_{\text{p}} m_{\text{e}}} \right) \left(1 + \sqrt{\frac{4}{D_{\text{D0}}}} \right) \mathfrak{E}$	$c = \left(\frac{L_{\text{p}}}{t_{\text{p}}} \right) \left(1 + \sqrt{G_{\text{G1}}} (\mathscr{K}_{\text{1}} - \mathscr{K}_{\text{2}}) \mathfrak{E} \right)$	$\text{F}_{\text{90}} = \text{farad} \left(1 + \frac{C_{\text{d}}}{6} (\mathscr{K}_{\text{1}} + \mathscr{K}_{\text{3}} + \mathscr{K}_{\text{4}}) \mathfrak{E} \right)$	$K_{\text{7}} = \frac{\mathscr{K}_{\text{1}}}{\pi} \left(\frac{t_{\text{p}} q_{\text{p}}}{t_{\text{p}}^2 m_{\text{p}}} \right) \left(1 - \frac{7}{8} \left(\frac{8}{4\pi} \right)^2 (\mathscr{K}_{\text{1}}^2 + \mathscr{K}_{\text{2}}^2) \mathfrak{E} \right)$	$\delta_{\text{de}} = \frac{2\pi}{16} \left(1 + \frac{16}{P_{\text{up}}} (\mathscr{K}_{\text{1}}^2 + \mathscr{K}_{\text{3}}^2 + \mathscr{K}_{\text{4}}^2) \mathfrak{E} \right)$	$t_{\text{p}} = 3.39125836832313... \times 10^{-35} \text{ s}$ $t_{\text{p}} = 1.61625918175645... \times 10^{-18} \text{ C}$ $q_{\text{p}} = 1.87554596713962... \times 10^{-18} \text{ C}$ $T_{\text{p}} = 1.41678698590795... \times 10^{32} \text{ K}$ $m_{\text{p}} = 2.17642683817579... \times 10^{-18} \text{ kg}$	Planck time Planck length Planck charge Planck temperature Planck mass	
$\text{K} \colon E_{\text{h}} = \frac{\text{kelvin}}{\mathscr{K}_1^4} \left(\frac{m_{\text{p}}}{T_{\text{p}} m_{\text{e}}} \right) \left(1 + 14 C_{\text{U}}^{-4} \mathscr{K}_{\text{p}} \mathfrak{E} \right)$	$m_{\text{p}} = \frac{1}{\mathscr{K}_{\text{3}} \mathscr{K}_{\text{4}}} \sqrt{\frac{16}{4\pi}} \left(m_{\text{n}} + m_{\text{e}} \right) \left(1 + \left(\frac{16}{2\pi} \right) \mathscr{K}_{\text{e}}^{-3} \mathfrak{E} \right)$	$R_{\text{K}} = \frac{2\pi}{\mathscr{K}_1^2} \left(\frac{t_{\text{p}}^2 m_{\text{p}}}{t_{\text{p}} q_{\text{p}}^2} \right) \left(1 + \frac{1}{2} Re \left(i^{i'''} \right) \mathscr{K}_{\text{3}} \mathscr{K}_{\text{4}} \mathfrak{E} \right)$	$Z_{\text{0}} = 4\pi \left(\frac{t_{\text{p}}^2 m_{\text{p}}}{t_{\text{p}}^2} \right) \left(1 + \frac{1}{2} Re \left(i^{i'''} \right) \mathscr{K}_{\text{3}} \mathscr{K}_{\text{4}} \mathfrak{E} \right)$	$\frac{\mu_{\text{N}}}{hc} = \frac{\mathscr{K}_{\text{1}}}{4\pi} \left(\frac{t_{\text{p}} q_{\text{p}}}{t_{\text{p}} m_{\text{e}}} \right) \left(1 + \sqrt{\frac{4}{D_{\text{D0}}}} \right) \mathfrak{E}$	$\text{H}_{\text{90}} = \text{henry} \left(1 - \frac{C_{\text{d}}}{6} (\mathscr{K}_{\text{1}} + \mathscr{K}_{\text{3}} + \mathscr{K}_{\text{4}}) \mathfrak{E} \right)$	$\Phi_{\text{0}} = \frac{\pi}{\mathscr{K}_1} \left(\frac{t_{\text{p}}^2 m_{\text{p}}}{t_{\text{p}} q_{\text{p}}} \right) \left(1 + \frac{7}{8} \left(\frac{8}{4\pi} \right)^2 (\mathscr{K}_{\text{1}}^2 + \mathscr{K}_{\text{2}}^2) \mathfrak{E} \right)$	$\frac{\mu_{\text{de}}}{\mu_{\text{p}}} = \frac{2\pi}{8\text{K}} \left(1 + \frac{16}{P_{\text{up}}} (\mathscr{K}_{\text{1}}^2 + \mathscr{K}_{\text{3}}^2 + \mathscr{K}_{\text{4}}^2) \mathfrak{E} \right)$	$\frac{\mu_{\text{de}}}{\mu_{\text{p}}} = \frac{2\pi}{8\text{K}} \left(1 + \frac{16}{P_{\text{up}}} (\mathscr{K}_{\text{1}}^2 + \mathscr{K}_{\text{3}}^2 + \mathscr{K}_{\text{4}}^2) \mathfrak{E} \right)$	$\mathfrak{E} = 0.999999199973626... 10^{-7}$ 5 7	hyperbolic inversion boundary number of dimensions break in scale symmetry
$\Delta_{\text{vCk}} = \frac{E_{\text{1}}}{2\pi} \left(\frac{t_{\text{p}}}{t_{\text{p}}^2 m_{\text{p}}} \right) \left(1 - \sqrt{\zeta(2)} \mathscr{K}_{\text{p}}^2 \mathfrak{E} \right)$	$m_{\text{e}} = \frac{2\pi}{\sqrt{\mathscr{K}_{\text{3}} \mathscr{K}_{\text{4}}}} \left(\frac{\text{GeV}}{\sqrt{8}} \right) \left(\frac{t_{\text{p}}^2}{t_{\text{p}}^2} \right) \left(1 + \frac{2\pi}{6^{1/3}} \left(\frac{2\pi}{2\text{s}} \right) \mathscr{K}_{\text{e}}^{-3} \mathfrak{E} \right)$	$Z_{\text{0}} = 4\pi \left(\frac{t_{\text{p}}^2 m_{\text{p}}}{t_{\text{p}}^2} \right) \left(1 + \frac{1}{2} Re \left(i^{i'''} \right) \mathscr{K}_{\text{3}} \mathscr{K}_{\text{4}} \mathfrak{E} \right)$	$Z_{\text{p}} = \left(\frac{t_{\text{p}}^2 m_{\text{p}}}{t_{\text{p}}^2} \right) \left(1 + \frac{1}{2} Re \left(i^{i'''} \right) \mathscr{K}_{\text{3}} \mathscr{K}_{\text{4}} \mathfrak{E} \right)$	$A_{\text{90}} = \text{ampere} \left(1 + \sqrt{\frac{9}{j_{\text{0.1}}}} \mathscr{K}_{\text{1}} \mathscr{K}_{\text{3}} \mathscr{K}_{\text{4}} \mathfrak{E} \right)$	$\Omega_{\text{00}} = \text{ohm} \left(1 - \frac{C_{\text{d}}}{6} (\mathscr{K}_{\text{1}} + \mathscr{K}_{\text{3}} + \mathscr{K}_{\text{4}}) \mathfrak{E} \right)$	$\frac{h}{e} = \frac{\text{coulomb}}{\mathscr{K}_1} \left(\frac{t_{\text{p}}^2 m_{\text{p}}}{t_{\text{p}} q_{\text{p}}} \right) \left(1 + \frac{7}{8} \left(\frac{8}{4\pi} \right)^2 (\mathscr{K}_{\text{1}}^2 + \mathscr{K}_{\text{2}}^2) \mathfrak{E} \right)$	$\frac{h}{e} = 2\pi \frac{\text{coulomb}}{\mathscr{K}_1} \left(\frac{t_{\text{p}}^2 m_{\text{p}}}{t_{\text{p}} q_{\text{p}}} \right) \left(1 + \frac{7}{8} \left(\frac{8}{4\pi} \right)^2 (\mathscr{K}_{\text{1}}^2 + \mathscr{K}_{\text{2}}^2) \mathfrak{E} \right)$	$c_{\text{1}} = (2\pi)^2 \left(\frac{t_{\text{p}}^4 m_{\text{p}}}{t_{\text{p}}^3} \right) \left(1 - \left(\frac{4\pi}{2} \right)^2 (\mathscr{K}_{\text{1}}^2 + \mathscr{K}_{\text{2}}^2) \mathfrak{E} \right)$	$\frac{b'}{s} = \frac{1}{2\pi} \left(\frac{v_{\text{peak}}}{t_{\text{p}} T_{\text{p}}} \right) \left(1 + \frac{18}{16} \left(\frac{2\pi}{s} \right)^2 (\mathscr{K}_{\text{1}}^2 - \mathscr{K}_{\text{3}}^2 - \mathscr{K}_{\text{4}}^2) \mathfrak{E} \right)$	$\mathscr{K}_{\text{1}} = 0.085425431533304... 1^{\text{st}}$ $\mathscr{K}_{\text{2}} = 3.667567535485501... 2^{\text{nd}}$ $\mathscr{K}_{\text{3}} = -1.87649603900417... + 4.06615262615972... i 3^{\text{rd}}$ $\mathscr{K}_{\text{4}} = -1.87649603900417... - 4.06615262615972... i 4^{\text{th}}$ $\mathscr{K}_{\text{p}} = 4.47826244916751... 1^{\text{st}}$ $\mathscr{K}_{\text{q}} = 2.00316562310924... 2^{\text{nd}}$
$\text{eV} \colon \text{Hz} = \frac{\text{eV}}{2\pi} \left(\frac{t_{\text{p}}}{t_{\text{p}}^2 m_{\text{p}}} \right) \left(1 - \sqrt{\zeta(2)} \mathscr{K}_{\text{p}}^2 \mathfrak{E} \right)$	$m_{\text{ne}} = \frac{\sqrt{3}}{\mathscr{K}_{\text{1}}^2 \mathscr{K}_{\text{2}}^2} \left(\frac{\text{GeV}}{2\pi} \right) \left(\frac{t_{\text{p}}^2}{t_{\text{p}}^2} \right) \left(1 - \frac{3^{-1}}{\mathscr{K}_{\text{1}} \mathscr{K}_{\text{2}}} \left(\frac{s}{18} \right) \mathscr{K}_{\text{e}}^{-3} \mathfrak{E} \right)$	$\text{Hz} \colon \frac$								